

CS-GY 6763: Lecture 3

Finishing Chebyshev's Inequality, Exponential Concentration Bounds

NYU, Prof. Ainesh Bakshi

DISTINCT ELEMENTS PROBLEM

Input: $x_1, \dots, x_n \in \mathcal{U}$ where $\mathcal{U}$ is a huge universe of items.

Output: Number of distinct inputs, D .

Example: $f(1, 10, 2, 4, 9, 2, 10, 4) \rightarrow D = 5$

Minimum-hash distinct-count sketch:

- Choose a fully random hash function $h : \mathcal{U} \rightarrow [0, 1]$ whose values on distinct inputs are independent and uniform.

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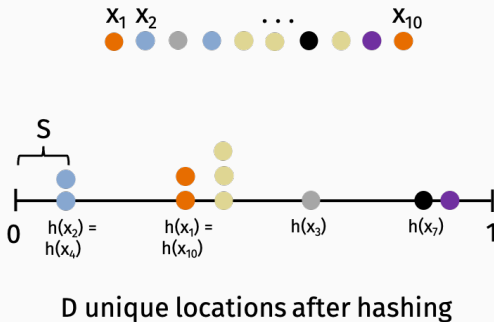
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- Return: $\frac{1}{S} - 1$

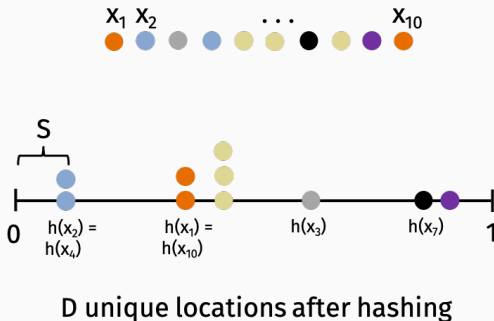
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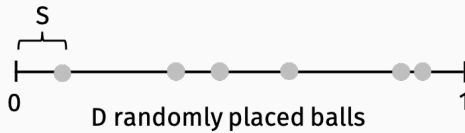


Intuition: When D is larger, S will be smaller. Makes sense to return the estimate $\tilde{D} = \frac{1}{S} - 1$.

What is $\mathbb{E}S$?

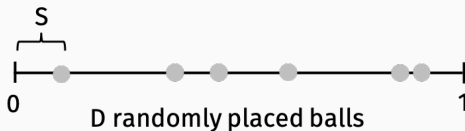
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Lemma

$$\mathbb{E}S = \frac{1}{D+1}.$$

THE CALCULUS PROOF

Proof:

$$\mathbb{E}[S] = \int_0^1 \Pr[S \geq \lambda] d\lambda$$

Exercise: Why?

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$$\begin{aligned}\mathbb{E}[S] &= \int_0^1 \Pr[S \geq \lambda] d\lambda \\ &= \int_0^1 (1 - \lambda)^D d\lambda\end{aligned}$$

Exercise: Why?

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Hint: For every non-negative random variable X , pointwise,
 $X = \int_0^\infty \mathbf{1}[X \geq t] dt$. Therefore,

$$\mathbb{E}[X] = \mathbb{E}\left[\int_0^\infty \mathbf{1}(X \geq t) dt\right] = \int_0^\infty \Pr[X \geq t] dt.$$

PROVING CONCENTRATION

Assume $D \geq 1$. Since $\mathbb{E}S = \frac{1}{D+1}$, use $\tilde{D} = \frac{1}{S} - 1$. For $0 < \epsilon < \frac{1}{4}$:

If $(1 - \epsilon)\mathbb{E}S \leq S \leq (1 + \epsilon)\mathbb{E}S$, then:

$$(1 - 4\epsilon)D \leq \tilde{D} \leq (1 + 4\epsilon)D.$$

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Lemma

$$\text{Var}[S] = \mathbb{E}[S^2] - \mathbb{E}[S]^2 = \frac{2}{(D+1)(D+2)} - \frac{1}{(D+1)^2} \leq \frac{1}{(D+1)^2}.$$

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www.wolframalpha.com/input?i=antiderivative+of+%281-sqrt%28x%29%29%5ED

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Recall we want to show that, with high probability,
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Chebyshev's: $\Pr[|S - \mu| \geq \epsilon\mu] \leq \Pr[|S - \mu| \geq \epsilon\sigma] \leq \frac{1}{\epsilon^2}$.

Vacuous bound. Our variance is way too high!

VARIANCE REDUCTION

Trick of the trade: Repeat many independent trials and take the mean to get a better estimator.

Given i.i.d. (independent, identically distributed) random variables $X_1, \dots, X_n$ with mean μ and variance σ^2 , what are:

- $\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n X_i \right] = \mu$

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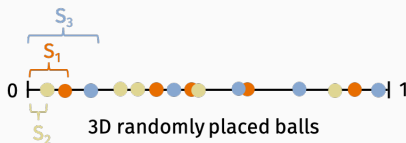
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Using independent hash functions, maintain k independent sketches $S_1, \dots, S_k$.

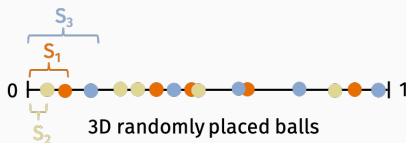


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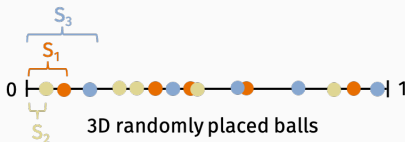


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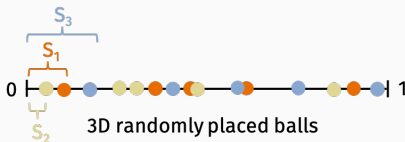


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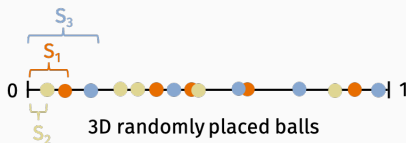


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Setting $c = 1/\sqrt{\delta}$ and $k = \lceil \frac{1}{\epsilon^2 \delta} \rceil$ gives:

$$\Pr[|S - \mu| \geq \epsilon\mu] \leq \delta.$$

Total space complexity: $O\left(\frac{1}{\epsilon^2 \delta}\right)$ to estimate distinct elements up to error ϵ with success probability $1 - \delta$.

NOTE ON FAILURE PROBABILITY

$O\left(\frac{1}{\epsilon^2 \delta}\right)$ space is an impressive bound:

- $1/\epsilon^2$ dependence cannot be improved.

¹Storing the sketch values to inverse-polynomial precision costs $O(k \log D)$ bits. The additional cost of describing the hash functions depends on the hash model.

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We can get a better bound depending on $O(\log(1/\delta))$ using exponential tail bounds. We will see this today.

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DISTINCT ELEMENTS IN PRACTICE

In practice, we cannot hash to real numbers on $[0, 1]$. Could use a finite grid, but more popular choice is to hash to integers (bit vectors).

Flajolet–Martin / HyperLogLog idea:

$h(x_1)$	10100 10
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- HyperLogLog keeps one maximum per bucket and combines the bucket estimates.

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Total Space: $O\left(\frac{\log \log D}{\epsilon^2} + \log D\right)$ for an ϵ approximate count.

“Using an auxiliary memory smaller than the size of this abstract, the LogLog algorithm makes it possible to estimate in a single pass and within a few percents the number of different words in the whole of Shakespeare’s works.” – Flajolet, Durand.

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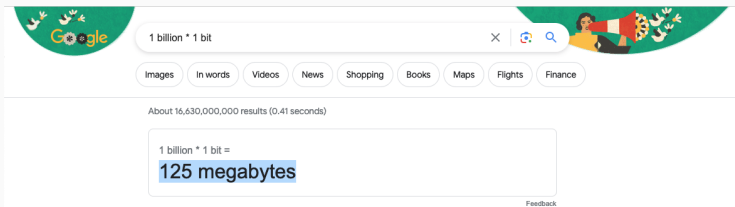
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$$m \geq \left(\frac{1.04}{\epsilon}\right)^2 = 2704; \quad \text{choose } m = 4096 \text{ registers,}$$

$$\begin{aligned} \text{space used} &\approx m \lceil \log_2 \log_2 D \rceil + \lceil \log_2 D \rceil \\ &= 4096 \cdot 5 + 30 = 20510 \text{ bits} \approx \text{2.50 KiB.} \end{aligned}$$

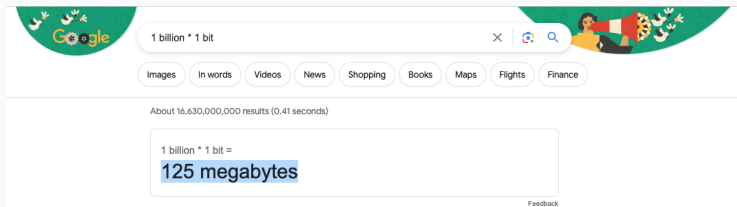
HYPERLOGLOG IN PRACTICE

For comparison, a bitmap over a known universe of 1 billion items takes 125 megabytes. Not tiny, but not unreasonable.



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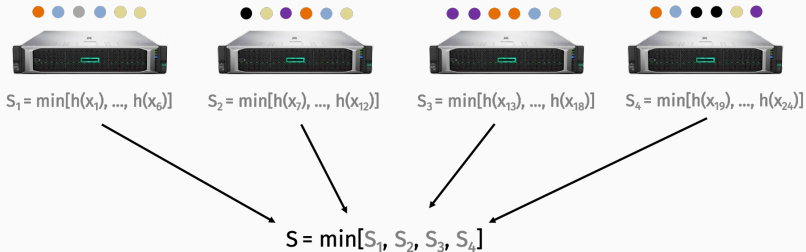
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These estimators become more important when you want to count many different things (e.g., a software company tracking clicks on hundreds of UI elements).

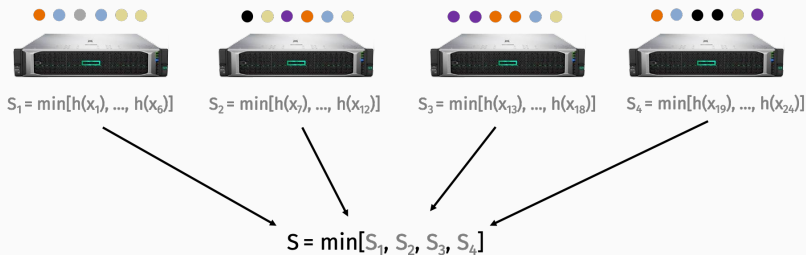
DISTRIBUTED DISTINCT ELEMENTS

Also very important in distributed settings.



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Distinct-elements summaries are “mergeable.” No need to share the underlying lists: merge minimum-hash sketches by taking the smaller minimum, and merge HyperLogLog sketches by taking register-wise maxima.

HYPERLOGLOG IN PRACTICE

Implementations: Google PowerDrill, Facebook Presto, Twitter Algebird, Amazon Redshift.

Use Case: Exploratory SQL-like queries on tables with hundreds of billions of rows.

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Answering a query requires a (distributed) linear scan over the database: 2 seconds in Google's distributed implementation.

Google Paper: "Processing a Trillion Cells per Mouse Click"

BEYOND CHEBYSHEV

Motivating question: Is Chebyshev's Inequality tight?

BEYOND CHEBYSHEV

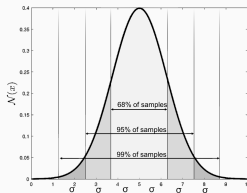
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68–95–99.7 rule for a Gaussian bell curve. $\mathbf{X} \sim \mathcal{N}(0, \sigma^2)$

Chebyshev's Inequality:

$$\Pr(|X - \mathbb{E}[X]| \geq 1\sigma) \leq 100\%$$

$$\Pr(|X - \mathbb{E}[X]| \geq 2\sigma) \leq 25\%$$

$$\Pr(|X - \mathbb{E}[X]| \geq 3\sigma) \leq 11.2\%$$

$$\Pr(|X - \mathbb{E}[X]| \geq 4\sigma) \leq 6.25\%.$$

Truth:

$$\Pr(|X - \mathbb{E}[X]| \geq 1\sigma) \approx 32\%$$

$$\Pr(|X - \mathbb{E}[X]| \geq 2\sigma) \approx 4.6\%$$

$$\Pr(|X - \mathbb{E}[X]| \geq 3\sigma) \approx 0.3\%$$

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GAUSSIAN CONCENTRATION

$X \sim \mathcal{N}(\mu, \sigma^2)$ has probability density function (PDF) p with:

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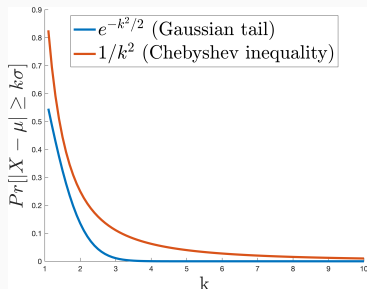
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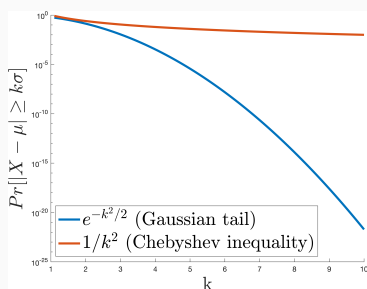
For $X \sim \mathcal{N}(\mu, \sigma^2)$ and $k > 0$:

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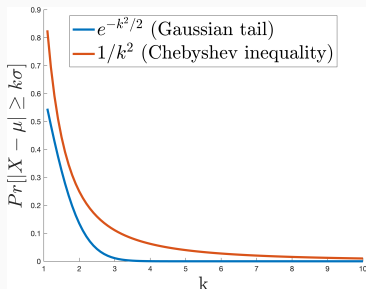


Standard y-scale.

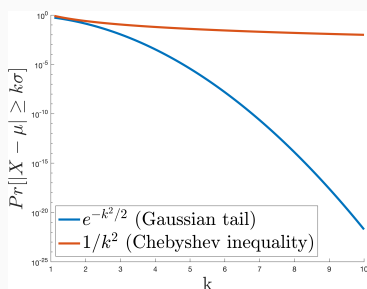


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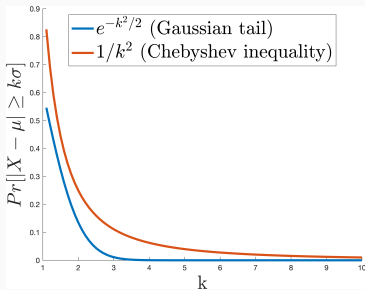
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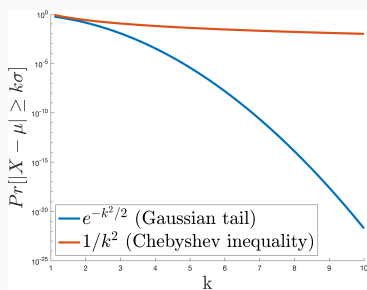
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Why does this matter for algorithm design?

CENTRAL LIMIT THEOREM

Theorem (CLT – Informal)

Let $X_1, X_2, \dots$ be i.i.d. random variables with mean μ and finite, positive variance σ^2 . If $S_n = \sum_{i=1}^n X_i$, then

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \implies \mathcal{N}(0, 1).$$

Equivalently, for large n , S_n is approximately $\mathcal{N}(n\mu, n\sigma^2)$.

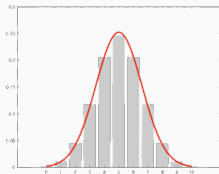
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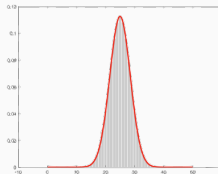
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(a) Distribution of # of heads after 10 coin flips, compared to a Gaussian.



(b) Distribution of # of heads after 50 coin flips, compared to a Gaussian.

INDEPENDENCE

Recall:

Definition (Mutual Independence)

Discrete random variables $X_1, \dots, X_n$ are mutually independent if, for all possible values $v_1, \dots, v_n$,

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Strictly stronger than pairwise independence.

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$2e^{-8} \approx 0.07\%$. Chebyshev's inequality gave a bound of 6.25%.

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These back-of-the-envelope calculations can be made rigorous! **Many different concentration bounds do so.**

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Different assumptions on random variables (e.g., binary vs. bounded), different forms (additive vs. multiplicative error), etc.

Wikipedia is your friend.

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Theorem (Chernoff Bound)

Let $X_1, X_2, \dots, X_n$ be independent $\{0, 1\}$ -valued random variables and let $p_i = \mathbb{E}[X_i]$, where $0 < p_i < 1$. Then $S = \sum_{i=1}^n X_i$, with mean $\mu = \mathbb{E}[S] = \sum_{i=1}^n p_i$, satisfies, for $\epsilon > 0$,

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When $\sigma^2 = \Theta(\mu)$, this has the same exponential dependence on the number of standard deviations, up to constants.

EXPONENTIAL CONCENTRATION INEQUALITIES

Theorem (Bernstein Inequality)

Let $X_1, X_2, \dots, X_n$ be independent random variables with each $X_i \in [-1, 1]$. Let $\mu_i = \mathbb{E}[X_i]$ and $\sigma_i^2 = \text{Var}[X_i]$. Let $\mu = \sum_{i=1}^n \mu_i$ and $\sigma^2 = \sum_{i=1}^n \sigma_i^2$. Then, for $0 < k \leq \frac{1}{2}\sigma$, $S = \sum_{i=1}^n X_i$ satisfies

$$\Pr[|S - \mu| > k \cdot \sigma] \leq 2e^{-k^2/4}.$$

EXPONENTIAL CONCENTRATION INEQUALITIES

Theorem (Hoeffding Inequality)

Let $X_1, X_2, \dots, X_n$ be independent random variables with each $X_i \in [a_i, b_i]$. Let $\mu_i = \mathbb{E}[X_i]$ and $\mu = \sum_{i=1}^n \mu_i$. Then, for any $k > 0$, $S = \sum_{i=1}^n X_i$ satisfies:

$$\Pr[|S - \mu| > k] \leq 2e^{\frac{-2k^2}{\sum_{i=1}^n (b_i - a_i)^2}}.$$

HOW ARE THESE BOUNDS PROVEN?

Variance is a natural measure of spread, but there are others. For even q :

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Idea in brief: Apply Markov's inequality to the non-negative random variable $|X - \mathbb{E}X|^q$ for larger even q , or to $\exp(\lambda(X - \mathbb{E}X))$. This requires control of higher moments or the moment-generating function, together with enough independence.

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Proof: Recall, Chernoff bound states

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We want $\alpha bn = \epsilon n$, so $\alpha = \epsilon/b$. Then, we have,

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Choosing $n \geq \frac{3 \ln(2/\delta)}{\epsilon^2}$ implies $2e^{-\epsilon^2 n/3} \leq \delta$.

LOAD BALANCING

Load balancing problem:

Suppose Google answers map search queries using servers $A_1, \dots, A_q$. Given a query like “new york to rhode island”, common practice is to choose a random hash function $h : \mathcal{U} \rightarrow [q]$ and to route this query to server:

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Why use a hash function instead of just distributing requests randomly?

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Let $x_1, \dots, x_m$ be distinct request IDs, and let s_i be the number assigned to server $i \in [n]$:

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Formally, our goal is to understand the value of maximum load on any server, which can be written as the random variable:

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If we distribute requests in round-robin fashion, $S = \lceil \frac{m}{n} \rceil$. But we have to repeat work.

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A good first step is to think about expectations. If we have n servers and m requests, and a uniformly random hash function, for any $i \in \{1, \dots, n\}$, what is $\mathbb{E}[s_i]$?

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Exercise: Give an example of two independent random variables A and B for which $\mathbb{E}[\max(A, B)] \neq \max(\mathbb{E}[A], \mathbb{E}[B])$.

SIMPLIFYING ASSUMPTIONS

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for as small a value of C as possible—ideally, much better than $C = n$.

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Goal: Prove that for some C ,

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$$\begin{aligned}\Pr[\max_i s_i \geq C] &\leq \sum_{i=1}^n \Pr[s_i \geq C] && \text{(Union bound)} \\ &\leq \sum_{i=1}^n \frac{1}{10n} = \frac{1}{10}. \quad \square\end{aligned}$$

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Let's try doing this with Markov's, Chebyshev, and exponential concentration.

ATTEMPT WITH MARKOV'S INEQUALITY

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- **Step 1.** Verify we can apply Markov's: s_i takes on non-negative values only. Good to go!

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Meaningless! There are only n balls, so of course there can't be more than $10n$ in the most overloaded bin.

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Let $s_{i,j}$ be a $\{0, 1\}$ indicator random variable for the event that ball j falls in bin i . We have:

$$s_i = \sum_{j=1}^n s_{i,j} = \sum_{j=1}^n \mathbb{1}[\text{ball } j \text{ falls in bin } i].$$

VARIANCE ANALYSIS

$$s_{i,j} = \begin{cases} 1 & \text{with probability } \frac{1}{n}, \\ 0 & \text{otherwise.} \end{cases}$$

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So with probability at least 90%, $S = \max_{i \in [n]} s_i \leq \sqrt{10n} + 1$.

FINAL RESULT FOR CHEBYSHEV'S

When hashing n balls into n bins, the maximum bin contains $O(\sqrt{n})$ balls with probability at least $\frac{9}{10}$.



Much better than the trivial bound of $n!$

ATTEMPT WITH EXPONENTIAL CONCENTRATION

Goal: Prove that $\Pr[s_i \geq C] \leq \frac{1}{10n}$.

Recall: $s_i = \sum_{j=1}^n s_{i,j}$, where $s_{i,j} = \mathbb{1}[\text{ball } j \text{ lands in bin } i]$.

What bound might we use?

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Chernoff bound!

Theorem (Chernoff Bound)

Let $X_1, X_2, \dots, X_n$ be independent $\{0, 1\}$ -valued random variables and let $p_j = \mathbb{E}[X_j]$, where $0 < p_j < 1$. Then $S = \sum_{j=1}^n X_j$, with mean $\mu = \sum_{j=1}^n p_j$, satisfies

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Apply with $S = s_i$, $X_j = s_{i,j}$, and $\mu = 1$. Set $\epsilon = c \ln n$.

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So, with probability at least $9/10$, this argument gives a maximum load of $O(\log n)$! Best we could prove with Chebyshev's was $O(\sqrt{n})$.

POWER OF TWO CHOICES

Power of 2 Choices: Instead of assigning a job to one random server, choose two random servers and assign it to the less loaded one. With probability at least $9/10$, the maximum load is bounded by:

- (a) $O(\log n)$ (b) $O(\sqrt{\log n})$ (c) $O(\log \log n)$ (d) $O(1)$

